# Customer Churn & Retention Analysis

## 1. Project Overview

This project analyzes customer churn behavior using telecom customer data (~7,000 records).

The primary objective is to identify churn drivers, measure revenue loss, segment high-risk customers, and develop data-driven retention strategies to improve long-term profitability.

By combining **Python (data preparation), MySQL (business analysis), and Power BI (visualization)**, this project delivers actionable insights for management decision-making.

## 2. Dataset Summary

- Rows: ~7,000

- Columns: 21

- Key Features:

- Customer demographics (Gender, SeniorCitizen, Partner, Dependents)

- Service details (PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies)

- Contract & billing information (Contract Type, PaperlessBilling, PaymentMethod, Tenure)

- Financial details (MonthlyCharges, TotalCharges)

- Target variable (Churn – Yes/No converted to 1/0)

- Missing Data:

  TotalCharges contained missing / blank values (converted to numeric and imputed using median)

## 3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported dataset using pandas.

- **Initial Exploration :** Used **df.info()** to inspect structure **df.describe()** for summary statistics **df.isnull().sum()** for missing values.

| | customerID | gender | SeniorCitizen | Partner | Dependents | tenure | PhoneService | MultipleLines | InternetService | OnlineSecurity | ... | DeviceProtection | TechSupport | StreamingTV | StreamingMovies | Contract | PaperlessBilling | PaymentMetho |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 7590-VHVEG | Female | 0 | Yes | No | 1 | No | No phone service | DSL | No | ... | No | No | No | No | Month-to-month | Yes | Electronic chec |
| 1 | 5575-GNVDE | Male | 0 | No | No | 34 | Yes | No | DSL | Yes | ... | Yes | No | No | No | One year | No | Mailed chec |
| 2 | 3668-QPYBK | Male | 0 | No | No | 2 | Yes | No | DSL | Yes | ... | No | No | No | No | Month-to-month | Yes | Mailed chec |
| 3 | 7795-CFOCW | Male | 0 | No | No | 45 | No | No phone service | DSL | Yes | ... | Yes | Yes | No | No | One year | No | Bank transfe (automati |
| 4 | 9237-HQITU | Female | 0 | No | No | 2 | Yes | No | Fiber optic | No | ... | No | No | No | No | Month-to-month | Yes | Electronic chec |

5 rows × 21 columns

| PaymentMethod | MonthlyCharges | TotalCharges | Churn |
|---|---|---|---|
| Electronic check | 29.85 | 29.85 | No |
| Mailed check | 56.95 | 1889.5 | No |
| Mailed check | 53.85 | 108.15 | Yes |
| Bank transfer (automatic) | 42.30 | 1840.75 | No |
| Electronic check | 70.70 | 151.65 | Yes |

**Data Cleaning :**

- Converted **TotalCharges** to **numeric**

- Replaced missing values with median

- Encoded Churn (Yes=1, No=0)

**Feature Engineering :**

- Created tenure_group (0–12, 12–24, 24–48, 48+)

- Calculated churn rate

- Calculated retention rate

- Estimated total revenue lost

**Key Python Insights :**

- Overall churn rate ≈ 26

- High churn observed in month-to-month contracts

- Customers with lower tenure show higher churn & High monthly charges correlate with churn

## 4. Data Analysis using MySQL (Business KPI Layer) :

Structured SQL analysis was performed to answer key business questions:

**1. Overall Churn Rate**

Calculated total customers, churned customers, churn rate %, and retention rate.

| | total_customers | churned_customers | churn_rate_percentage |
|---|---|---|---|
| ▶ | 7043 | 1869 | 26.54 |

**2. Churn by Contract Type**

Identified highest churn in month-to-month contracts.

| | Contract | total_customers | churned_customers | churn_rate |
|---|---|---|---|---|
| ▶ | Month-to-month | 3875 | 1655 | 42.71 |
| | One year | 1473 | 166 | 11.27 |
| | Two year | 1695 | 48 | 2.83 |

**3. Revenue Lost Due to Churn**

Computed total revenue lost from churned customers using TotalCharges.

| | monthly_revenue_lost |
|---|---|
| ▶ | 139130.85 |

**4. Churn by Tenure Group**

Segmented customers based on tenure and calculated churn rate per segment.

| | tenure_group | total_customers | churned_customers | churn_rate |
|---|---|---|---|---|
| ▶ | 0-12 Months | 2186 | 1037 | 47.44 |
| | 13-24 Months | 1024 | 294 | 28.71 |
| | 24+ Months | 3833 | 538 | 14.04 |

## 5. Customer Spending Ranking

Used SQL window functions (RANK, NTILE) to identify high-value customers.

| customerID | MonthlyCharges | charge_rank |
|---|---|---|
| 7569-NMZYQ | 118.75 | 1 |
| 8984-HPEMB | 118.65 | 2 |
| 5989-AXPUC | 118.60 | 3 |
| 5734-EJKXG | 118.60 | 3 |
| 8199-ZLLSA | 118.35 | 5 |
| 9924-JPRMC | 118.20 | 6 |
| 2889-FPWRM | 117.80 | 7 |
| 3810-DVDQQ | 117.60 | 8 |
| 9739-JLPQJ | 117.50 | 9 |
| 2302-ANTDP | 117.45 | 10 |
| 6904-JLBGY | 117.35 | 11 |
| 4282-MSACW | 117.20 | 12 |
| 6650-BWFRT | 117.15 | 13 |
| 9788-HNGUT | 116.95 | 14 |
| 1488-PBLJN | 116.85 | 15 |
| 0017-IUDMW | 116.80 | 16 |
| 8628-MFKAX | 116.75 | 17 |
| 3680-CTHUH | 116.60 | 18 |
| 3258-ZKPAI | 116.60 | 18 |
| 3795-CAWEX | 116.55 | 20 |
| 8016-NCFVO | 116.50 | 21 |
| 3963-RYFNS | 116.45 | 22 |
| 0675-NCDYU | 116.40 | 23 |
| 0186-CAERR | 116.30 | 24 |
| 4374-YMLISO | 116.25 | 25 |

| customerID | MonthlyCharges | charge_rank |
|---|---|---|
| 4635-EJYPD | 18.95 | 7018 |
| 7968-QUXNS | 18.95 | 7018 |
| 9025-ZRPVR | 18.95 | 7018 |
| 1386-ZIKUV | 18.95 | 7018 |
| 3719-TDVQB | 18.95 | 7018 |
| 9975-SKRNR | 18.90 | 7024 |
| 5025-GOOKI | 18.90 | 7024 |
| 8992-CEUEN | 18.85 | 7026 |
| 0620-XEFWH | 18.85 | 7026 |
| 5863-OOKCL | 18.85 | 7026 |
| 3572-UUHRS | 18.85 | 7026 |
| 9318-NKNFC | 18.85 | 7026 |
| 2967-MXRAV | 18.80 | 7031 |
| 7369-TRPFD | 18.80 | 7031 |
| 7473-ZBDSN | 18.80 | 7031 |
| 3806-YAZOV | 18.80 | 7031 |
| 8464-EETCQ | 18.80 | 7031 |
| 3387-PLKUI | 18.80 | 7031 |
| 6508-NJYRO | 18.80 | 7031 |
| 9426-SXNHE | 18.75 | 7038 |
| 0621-CXBKL | 18.70 | 7039 |
| 9945-PSVIP | 18.70 | 7039 |
| 0827-ITJPH | 18.55 | 7041 |
| 9764-REAFF | 18.40 | 7042 |
| 6823-SIDFQ | 18.25 | 7043 |

## 6. Revenue Lost by Contract

Measured financial impact per contract category.

| CONTRACT | CHURNED_CUSTOMERS | REVENUE_LOST |
|---|---|---|
| Month-to-month | 1655 | 1927182.25 |
| One year | 166 | 674991.20 |
| Two year | 48 | 260753.45 |

This analysis connected churn behavior directly to revenue impact.

## 5. Dashboard in Power BI

An interactive executive dashboard was built in Power BI to visualize churn metrics.

**Dashboard KPIs:**

- Total Customers

- Churn Rate %

- Retention Rate %

- Total Revenue

- Revenue Lost

**Visual Analysis Sections:**

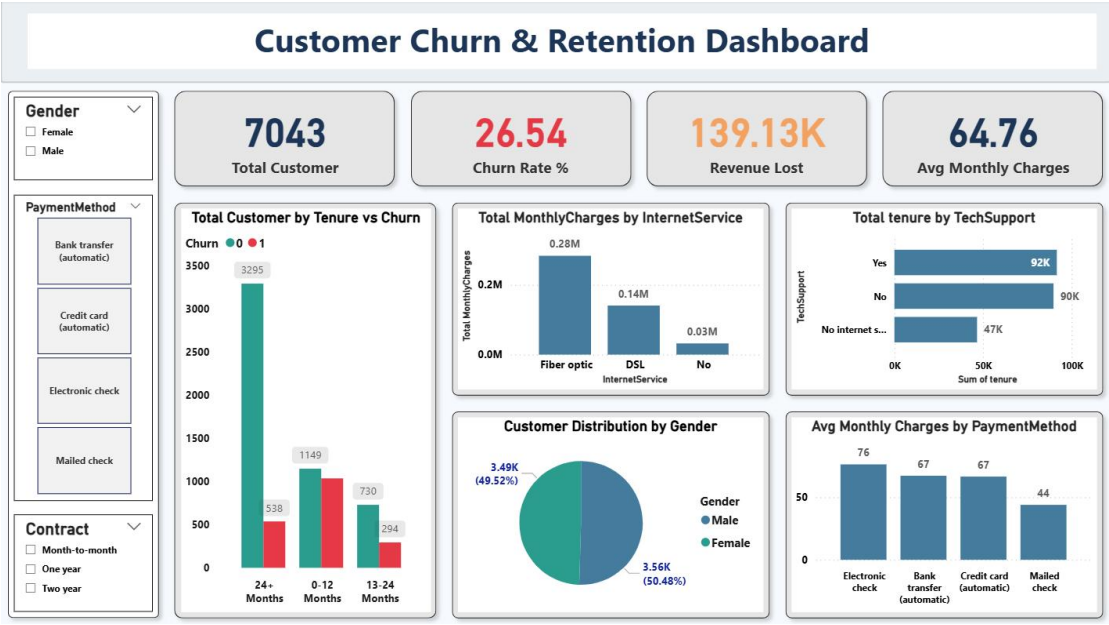
- Churn by Contract Type & Tenure Group

- Monthly Charges Distribution

- Revenue Lost by Segment

- High-Risk Customer Segmentation

- Interactive filters allow stakeholders to explore customer segments dynamically.



## 6. Business Recommendations

● Promote long-term contracts to reduce month-to-month churn.

● Improve onboarding experience during first 12 months.

● Offer targeted discounts to high monthly charge customers.

● Introduce loyalty programs for long-tenure customers.

● Implement predictive churn monitoring to proactively retain high-risk customers.

## Conclusion :

*This project demonstrates an end-to-end data analytics workflow:*

- Data cleaning and transformation using Python

- KPI and revenue analysis using SQL

- Interactive dashboard development in Power BI

- Strategic business recommendations

- The insights generated can help reduce churn, improve retention, and enhance long-term profitability.